Chapter Outline: Valuing Stocks

1. The Dividend Discount Model
2. Applying the Dividend Discount Model

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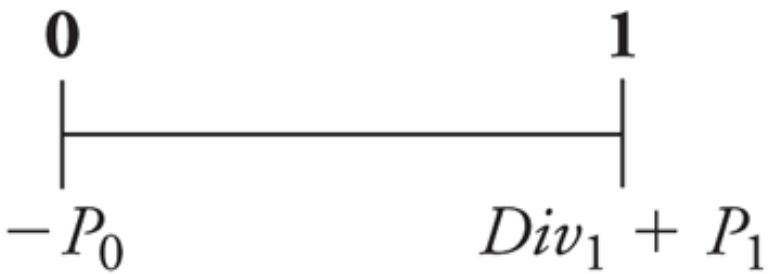
1. The Dividend Discount Model

- A One-Year Investor
  - Potential Cash Flows
    - Dividend
    - Sale of Stock
  - Timeline for One-Year Investor



    - Since the cash flows are risky, we must discount them at the **equity cost of capital.**

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1. The Dividend Discount Model (cont'd)

- A One-Year Investor

$$P_0 = \left(\frac{Div_1 + P_1}{1 + r_E} \right)$$

  - If the current stock price were less than this amount, expect investors to rush in and buy it, driving up the stock's price.
  - If the stock price exceeded this amount, selling it would cause the stock price to quickly fall.

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Dividend Yields, Capital Gains, and Total Returns

$$r_E = \frac{Div_1 + P_1}{P_0} - 1 = \underbrace{\frac{Div_1}{P_0}}_{\text{Dividend Yield}} + \frac{P_1 - P_0}{\underbrace{P_0}_{\text{Capital Gain Rate}}}$$

- Dividend Yield
- Capital Gain
  - Capital Gain Rate
- Total Return
  - Dividend Yield + Capital Gain Rate
    - *The expected total return of the stock should equal the expected return of other investments available in the market with equivalent risk.*

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Textbook Example 1

Stock Prices and Returns

Problem

Suppose you expect Walgreen Company (a drugstore chain) to pay dividends of $0.44 per share and trade for $33 per share at the end of the year. If investments with equivalent risk to Walgreen's stock have an expected return of 8.5%, what is the most you would pay today for Walgreen's stock? What dividend yield and capital gain rate would you expect at this price?

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Textbook Example 1 (cont'd)

Solution

Using Eq. 9.1, we have

$$P_0 = \frac{Div_1 + P_1}{1 + r_E} = \frac{0.44 + 33.00}{1.085} = \$30.82$$

At this price, Walgreen's dividend yield is $Div_1/P_0 = 0.44/30.82 = 1.43\%$. The expected capital gain is $\$33.00 - \$30.82 = \$2.18$ per share, for a capital gain rate of $2.18/30.82 = 7.07\%$. Therefore, at this price, Walgreen's expected total return is $1.43\% + 7.07\% = 8.5\%$, which is equal to its equity cost of capital.

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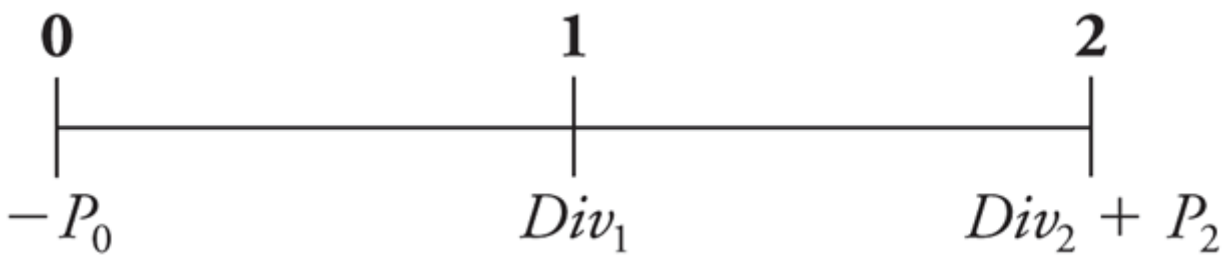
A Multi-Year Investor

- What is the price if we plan on holding the stock for two years?



$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2 + P_2}{(1 + r_E)^2}$$

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The Dividend-Discount Model Equation

- What is the price if we plan on holding the stock for N years?

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \cdots + \frac{Div_N}{(1 + r_E)^N} + \frac{P_N}{(1 + r_E)^N}$$

  - This is known as the Dividend Discount Model.
    - Note that the above equation (9.4) holds for any horizon *N*. Thus all investors (with the same beliefs) will attach the same value to the stock, independent of their investment horizons.

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The Dividend-Discount Model Equation (cont'd)

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \frac{Div_3}{(1 + r_E)^3} + \cdots = \sum_{n=1}^{\infty} \frac{Div_n}{(1 + r_E)^n}$$

- *The price of any stock is equal to the present value of the expected future dividends it will pay.*

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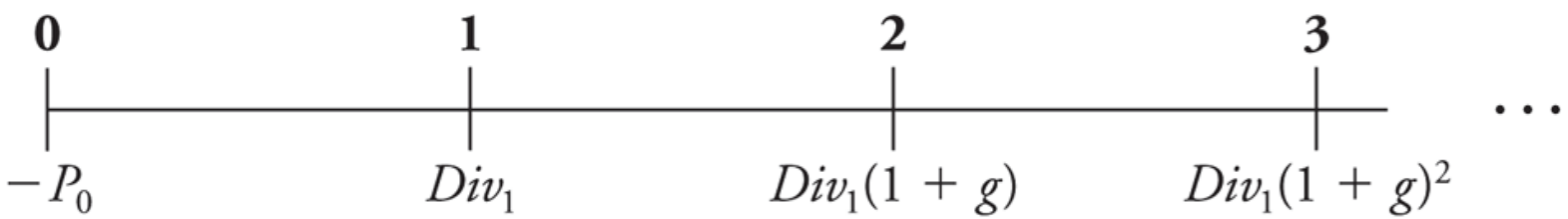
2. Applying the Discount-Dividend Model

- Constant Dividend Growth
  - The simplest forecast for the firm's future dividends states that they will grow at a constant rate, *g*, forever.



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2. Applying the Discount-Dividend Model (cont'd)

- Constant Dividend Growth Model

$$P_0 = \frac{Div_1}{r_E - g}$$

$$r_E = \frac{Div_1}{P_0} + g$$

  – The value of the firm depends on the current dividend level, the cost of equity, and the growth rate.

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Textbook Example 2

Valuing a Firm with Constant Dividend Growth

Problem

Consolidated Edison, Inc. (Con Edison), is a regulated utility company that services the New York City area. Suppose Con Edison plans to pay $2.36 per share in dividends in the coming year. If its equity cost of capital is 7.5% and dividends are expected to grow by 1.5% per year in the future, estimate the value of Con Edison's stock.

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Textbook Example 2 (cont'd)

Solution

If dividends are expected to grow perpetually at a rate of 1.5% per year, we can use Eq. 9.6 to calculate the price of a share of Con Edison stock:

$$P_0 = \frac{Div_1}{r_E - g} = \frac{\$2.36}{0.075 - 0.015} = \$39.33$$

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Changing Growth Rates

- We cannot use the constant dividend growth model to value a stock if the growth rate is not constant.
  - For example, young firms often have very high initial earnings growth rates. During this period of high growth, these firms often retain 100% of their earnings to exploit profitable investment opportunities. As they mature, their growth slows. At some point, their earnings exceed their investment needs and they begin to pay dividends.

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Changing Growth Rates (cont'd)

- Although we cannot use the constant dividend growth model directly when growth is not constant, we can use the general form of the model to value a firm by applying the constant growth model to calculate the future share price of the stock once the expected growth rate stabilizes.

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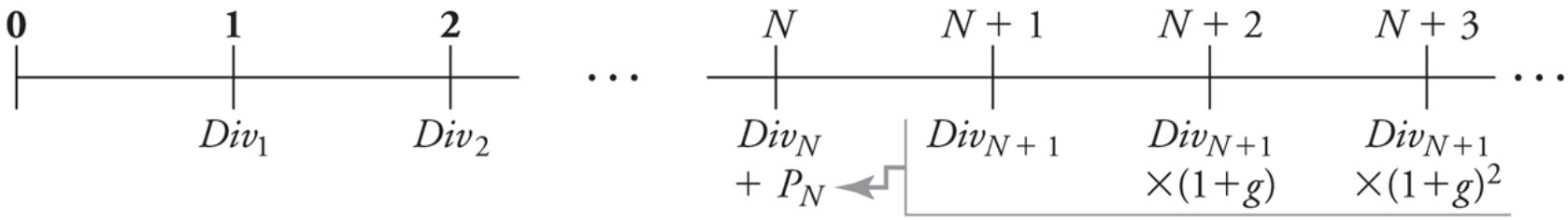
Changing Growth Rates (cont'd)



$$P_N = \frac{Div_{N+1}}{r_E - g}$$

- Dividend-Discount Model with Constant Long-Term Growth

$$P_0 = \frac{Div_1}{1 + r_E} + \frac{Div_2}{(1 + r_E)^2} + \cdots + \frac{Div_N}{(1 + r_E)^N} + \frac{1}{(1 + r_E)^N}\left(\frac{Div_{N+1}}{r_E - g}\right)$$

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